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AI for Bharat Hackathon

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Team Name : MandiMind

Team Leader Name : Chockalingam Rajendran

Problem Statement : AI for Rural Innovation & Sustainable Systems

Brief about the Idea:

The Rural Producer Intelligence Network (RPIN) is a modular, AI-powered platform providing intelligent market recommendations. It integrates historical mandi data, weather forecasts, and transport costs to predict prices and demand, helping producers choose the best time and place to sell.

What Makes RPIN Different, Effective, and Unique

- Most agri/rural solutions focus on crop advice, weather alerts, or showing current mandi prices
- RPIN combines price forecasting, demand prediction, spoilage risk, and transport cost to recommend where and when to sell before harvest
- RPIN removes guesswork and middlemen dependency by giving rural producers a clear, data-driven comparison of multiple markets with net profit calculation.Producers can choose the best market that increases income and reduces wastage.

USP of the proposed solution

- AI-powered market decision engine (not a dashboard)
- Future price + demand + spoilage + logistics in one system
- Net profit–based market ranking
- Simple natural-language explanation for every recommendation
- Works with public data and is scalable across villages, crops, and markets

RPIN helps producers decide the best market before harvest, not after loss.

Key Features of RPIN (Rural Producer Intelligence Network)

- Intelligent Producer Input
- AI Price Forecasting (Next 7 Days)
- Demand Level Prediction
- Spoilage Risk Assessment
- Transport Cost Estimation
- Net Profit Optimization (Core Feature)
- AI Explanation in Simple Language

Input Details

- Village
- Crop Type
- Quantity
- Harvest Date

Profit Comparison

	Price	Demand	Cost	Profit
Market	₹18	Medium	High	₹17,50
Market	₹22	High	₹17,500	₹17,500
Market	₹20	Low	Low	₹17,300

Rural Producer

AI-Powered Analysis

- Price Prediction
- Demand Prediction
- Spoilage Estimation
- Transport Cost Calculation

AI Recomendation

Madurai is best because price rising, high demand and low spoilage risk

Highest Profit Market & Selling Day Recommendation

Tells the producer **'why'** a market is

Converts analytics into easy decision

Recommended Market

Rural Producer

System Process Flow

Rural Producer

Input: Village, Crop, Quantity, Harvest Date

AI Backend

- Prediction Service
- Optimization Service
- Spoilage Estimation

Analyze Data with ML Models

Market	Price	Demand	Cost	Profit
Market A	₹18	Medium	High	₹17,50
Market B	₹22	High	₹17,500	₹17,50
Market C	₹20	Low	Low	₹17,50

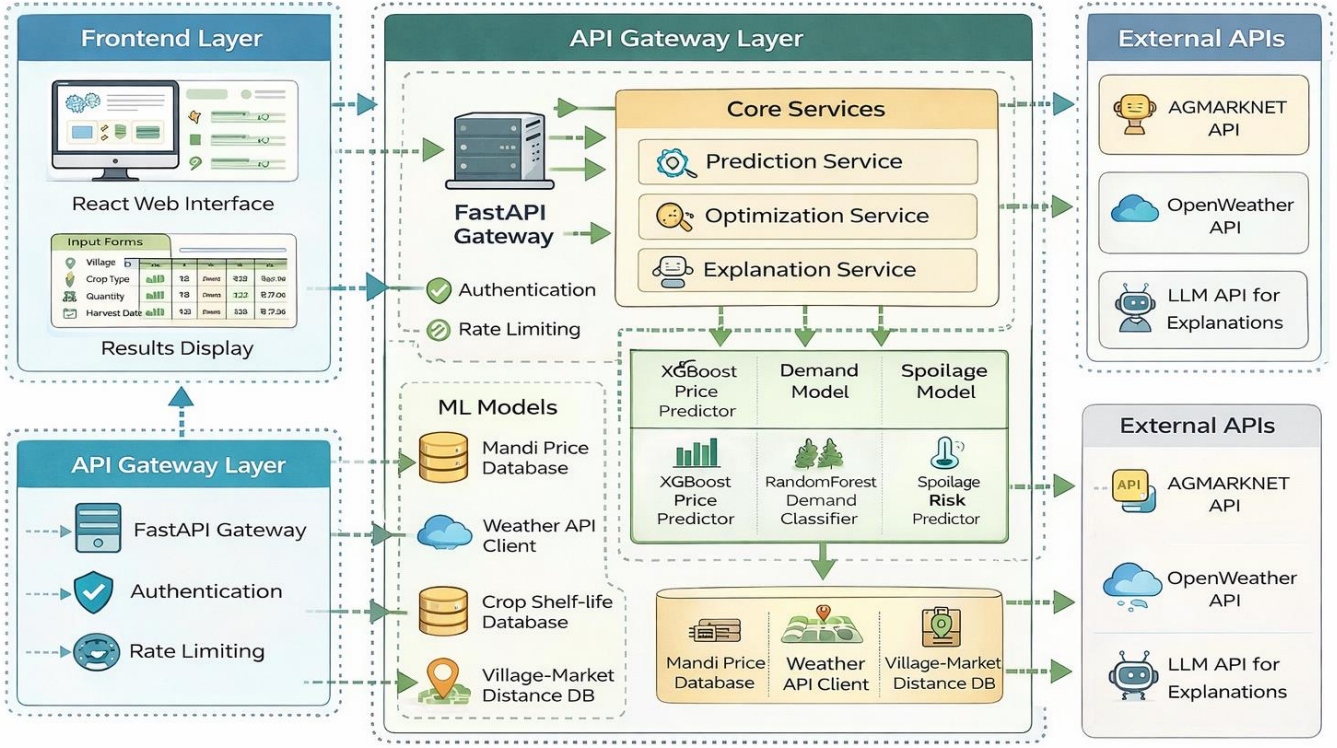
Generate Simple Recommendation

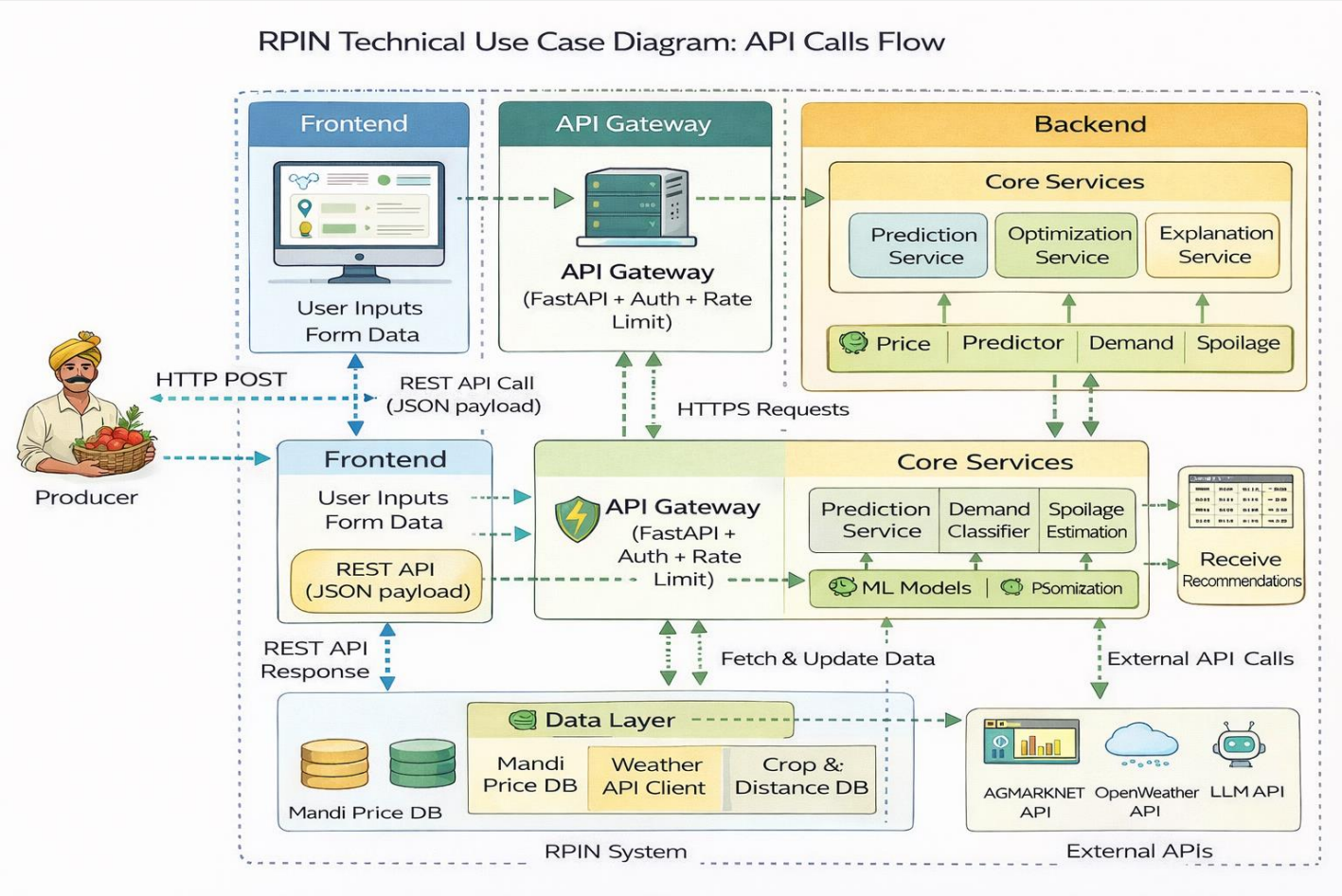
Madurai is best because price rising, high demand and low spoilage risk

Rural Producer

Provide Output to Producer

High-Level System Architecture




RPIN Technical Use Case Diagram: API Calls Flow
Frontend
User Inputs
Form Data
API Gateway
API Gateway
(FastAPI + Auth + Rate Limit)
Backend
Core Services
Prediction Service
Optimization Service
Explanation Service
Price
Predictor
Demand
Spoilage
HTTP POST
REST API Call
(JSON payload)
HTTPS Requests
Producer
Frontend
User Inputs
Form Data
REST API
(JSON payload)
API Gateway
(FastAPI + Auth + Rate Limit)
Core Services
Prediction Service
Demand Classifier
Spoilage Estimation
ML Models
PSomization
Receive
Recommendations
REST API
Response
Fetch & Update Data
External API Calls
Data Layer
Mandi Price DB
Weather API Client
Crop & Distance DB
Mandi Price DB
AGMARKNET API
OpenWeather API
LLM API
RPIN System
External APIs

Technologies to be used in the solution:

- Backend: FastAPI (Python 3.9+)
- Machine Learning: XGBoost, scikit-learn, pandas
- Frontend: React, TypeScript
- Database/Storage: SQLite, JSON configuration
- APIs: data.gov.in (AGMARKNET), OpenWeatherMap, LLM API
- Testing: Hypothesis (Property-based testing)

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Thank You